3. Part 2 — Evolution and refactoring

The changes evolve the existing ha-fusion application from baseline commit 65cc1957c39c6005609102c33650c32abc8d12f0. CH01 is commit b2731a34870bcb79090327234341e431b3c24b28; CH02 is commit 3eb21fc061b3aefa339ae9b9f92cd51f68c4b567. CH03 is commit 632f01bd326e09841d0279fbf5f3ca33c1e4940f on ch03-hearth-widget-extension, based on CH02. Its source files are also pinned by the SHA-256 manifest at Part2/code/snapshots/Change3_snapshot.sha256 and listed in Part2/code/snapshots/Change3_SHA256SUMS.

CH01 — Negation and condition evaluator dispatch

Requirement and acceptance. Add a recursive not condition that evaluates false for an empty or absent child list and otherwise negates whether any child evaluates true. Preserve existing state, numeric, screen, and and/or behavior. Requirements and observable acceptance criteria are in Part2/code/changes/Change1/requirement.md and acceptance_criteria.md.

Implementation and refactoring. Condition gained the not discriminant; handleNot() implements negation; and validateCondition() now selects evaluator functions from CONDITION_EVALUATORS. This replaces the condition-type switch with a function-based Strategy dispatch table. Changed paths are src/lib/Conditional.ts, src/lib/Types.ts, and new src/lib/Conditional.test.ts. The commit diff and verification logs are in Part2/code/changes/Change1/.

Pattern effect. The Visibility Condition Tree Composite is extended with not. The evaluator dispatch is recorded as a newly introduced, function-based Strategy instance. Other tracked collaborations are preserved.

Verification. CH01's saved run reports 14 files and 59 tests passing. After cumulative type-check corrections, the final saved run reports 15 files and 68 tests passing, with pnpm check reporting 0 errors and 0 warnings. See Part2/code/changes/Change1/test_result.log, data/test_result/evolved_test.log, and data/test_result/evolved_check.log.

Metrics. The same two files were measured before and after: Conditional.ts plus Types.ts. cloc source LOC changed from 468 to 484 (+16). Lizard CCN for validateCondition() changed from 6 to 3; new handleNot() has CCN 4. Raw outputs are data/test_result/ch01_cloc_{before,after}.log and ch01_lizard_{before,after}.csv.

CH02 — Hearth history caretaker extraction

Requirement and acceptance. Encapsulate Hearth configuration history in a generic HistoryManager<T>, preserve the store's observable undo/redo behavior, enforce a configurable history bound, and expose clear/depth operations. Requirements and criteria are in Part2/code/changes/Change2/.

Implementation and refactoring. src/lib/Hearth/historyManager.ts now owns bounded undo/redo

stacks, push, undo, redo, clear, availability, and depth operations. store.ts delegates to it. This applies Extract Class and Encapsulate Collection to scattered stack mutation. Changed paths are new src/lib/Hearth/historyManager.ts and historyManager.test.ts, plus src/lib/Hearth/store.ts. The CH02 commit diff is Part2/code/changes/Change2/change.diff.

Pattern effect. The Hearth Config History & Edit Transaction remains a Memento collaboration, with the new HistoryManager<HearthConfig> as its caretaker. The visibility Strategy from CH01 is preserved.

Verification. The saved CH02 run reports 15 test files and 67 tests passing. The final cumulative run after stabilization reports 68 passing tests and a clean type check. Logs are Part2/code/changes/Change2/test_result.log and the cumulative logs under data/test_result/.

Metrics. The same src/lib/Hearth/ directory was measured in the CH01 and CH02 snapshots. cloc source LOC changed from 17,063 to 17,186 (+123). Lizard's maximum CCN across the before/after history operations remained 2; the refactor's benefit is encapsulation and testability, not a measured CCN decrease. Raw outputs are data/test_result/ch02_cloc_{before,after}.log and ch02_lizard_{before,after}.csv.

CH03 – Alert rail widget

Requirement and acceptance. Add an alert widget variant and registry entry, require text or an entity for configuration, and render alert content with severity. The requirement and criteria are in Part2/code/changes/Change3/.

Implementation and refactoring. The alert variant is registered and rendered by a new AlertWidget.svelte. configurationState.ts dispatches per-type configuration predicates through WIDGET_CONFIG_VALIDATORS, replacing a growing conditional with a function-based Strategy map. ConfigurationPlaceholder.svelte and RailWidgetRenderer.svelte were updated for widget-specific labels. Changed paths are src/lib/Hearth/config.ts, configurationState.ts, configurationState.test.ts, RailWidgetRenderer.svelte, ConfigurationPlaceholder.svelte, new AlertWidget.svelte, and the check-driven corrections in src/lib/Types.ts, src/lib/Conditional.ts, and src/lib/Hearth/store.ts. The main CH03 diff is Part2/code/changes/Change3/change.diff; final source delta from the CH02 base is Part2/code/changes/Change3/final_source_delta.diff.

Pattern effect. The rail validator dispatch is a newly introduced function-based Strategy instance with the alert predicate. Existing tracked collaborations are preserved; the Hearth Memento has a final history-clear correction in saveEdit().

Verification. The saved final cumulative commands pnpm test and pnpm check report 15 test files / 68 tests passing and 0 errors / 0 warnings, respectively. The earlier CH03 check errors and their corrections are retained in

Part2/code/changes/Change3/test_result.log for transparency.

Metrics. The same src/lib/Hearth/ directory was measured in CH02 and the CH03 working-tree snapshot. cloc source LOC changed from 17,186 to 17,293 (+107). Lizard CCN for railWidgetNeedsConfiguration() changed from 7 to 2; the new alert predicate has CCN 2. Raw outputs are data/test_result/ch03_cloc_{before,after}.log and ch03_lizard_{before,after}.csv.

3.1 Updated pattern counts

Counts below refer to the tracked inventory in data/patterns_evolution.csv; they are not a claim that every pattern in the full application has been freshly audited. “Concrete participant classes” counts distinct source class symbols serving concrete roles: KonvaEditor, KonvaViewer, and, after CH02, HistoryManager. Strategy participants are functions, not classes.

Snapshot	Distinct tracked pattern types	Tracked instances	Concrete participant classes	Rejected observations recorded	Uncertain observations recorded
Original	5	8	2	0	0
After CH01	6	9	2	0	0
After CH02	6	9	3	0	0
After CH03	6	10	3	1	0

The CH03 change-scope review rejected one “Hearth Rail Widget Factory Method” candidate: the widget descriptor registry and renderer branches do not form a Factory Method collaboration. The observation is recorded in data/pattern_candidate_observations.csv. No uncertain rows are recorded. Baseline zeros mean no observations were retained in the carried-forward inventory; the independent Part 1 candidate audit was skipped, so they do not mean an exhaustive baseline sweep found none.

The change sequence adds recursive negation and lowers evaluator-dispatch CCN, then isolates history management without changing its measured maximum CCN, then adds an alert variant and lowers rail-configuration dispatch CCN. CH02 adds a class and CH03 adds a component and validators, so source LOC increases are expected; LOC alone does not establish design quality. The trade-offs are additional abstractions and tests to maintain, while behavior is protected by the passing cumulative suite. The remaining limitation is that the inherited baseline pattern inventory has not received an independent audit.

Evidence and reproducibility

The inventory and count dataset are data/patterns_evolution.csv and data/pattern_counts.csv. Metric summaries and raw logs are linked above. The updated evolution diagram source and rendering are Part2/diagrams/evolution_patterns.mmd and Part2/diagrams/evolution_patterns.svg. Snapshot provenance is documented in Part2/code/evolved_source_provenance.md and Part2/evolution_evidence_notes.md.